

Interpretable ECG Signal Analysis using Feature Based Machine Learning and SHAP Explainability

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ABSTRACT

This thesis presents a comprehensive study on real-time ECG signal analysis using machine learning and explainable AI. The primary objective is to develop an automated, web-based platform for accurate and transparent cardiac condition classification. We propose a novel approach integrating a configurable signal preprocessing pipeline with multi-model machine learning and SHAP-based feature explainability. The experimental results demonstrate that the system significantly improves signal quality, achieves high classification accuracy, and provides clear interpretability. This work contributes to the advancement of reliable and accessible clinical decision support systems for cardiovascular diagnostics.

Furthermore, the platform addresses the challenge of noisy, artifact-prone ECG signals by providing an automated preprocessing pipeline. Built with a React-based frontend, a Node.js backend, and a Python analytics engine, it supports real-time visualization and interactive control over denoising operations such as DC removal, baseline wander correction, and notch/bandpass filtering. Following preprocessing, clinically relevant features from 12-lead ECG signals are extracted and classified using models including Random Forest, XGBoost, LightGBM, and a Multi-Layer Perceptron (MLP), with XGBoost yielding the best performance. To enhance interpretability, SHapley Additive Explanations (SHAP) provide feature-level and lead-wise insights into model predictions. Ultimately, the system improves diagnostic accuracy while maintaining essential transparency and usability for both clinical and research applications.

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Chapter 1

Introduction

Cardiovascular diseases (CVDs) constitute the leading cause of mortality globally, precipitating an urgent demand for scalable, robust, and accessible diagnostic frameworks [15, 19]. The early and precise detection of cardiac anomalies is paramount for facilitating timely clinical interventions and optimizing patient prognoses. Electrocardiography (ECG) remains the prevailing non-invasive modality for assessing cardiac electrophysiology. By capturing the electrical depolarization and repolarization of the myocardium, ECG yields critical clinical insights through morphological components such as the P-wave, QRS complex, and T-wave [7].

Despite its pervasive clinical utility, the automated and manual interpretation of ECG signals is frequently compromised by environmental and physiological artifacts. In practical settings, raw ECG recordings are highly susceptible to baseline wander, power-line interference, electromyographic (EMG) noise, and electrode motion artifacts. These perturbations significantly degrade signal-to-noise ratios, obscuring critical morphological features and rendering raw data suboptimal for direct diagnostic interpretation [13]. Consequently, the implementation of robust, multi-stage signal preprocessing pipelines is an absolute prerequisite for accurate downstream analysis.

Historically, the computational analysis of ECG data has been relegated to offline, heavily

siloed clinical environments. These traditional paradigms often lack interoperability, real-time processing capabilities, and dynamic user interactivity. Furthermore, the advent of deep learning and complex ensemble models in cardiology has introduced a "black-box" dilemma. Many contemporary machine learning architectures prioritize predictive accuracy at the expense of clinical transparency, limiting their adoption in high-stakes healthcare environments where algorithmic interpretability is a strict requirement.

To bridge these critical gaps, this thesis proposes the architecture and deployment of a comprehensive, web-based ECG analysis platform. By seamlessly integrating signal processing algorithms, high-performance machine learning models, and Explainable Artificial Intelligence (XAI), the system delivers an end-to-end pipeline tailored for both clinical and research applications. The architecture employs a modern full-stack paradigm: a React-based frontend facilitates interactive, real-time data visualization; a Node.js backend orchestrates asynchronous data handling; and dedicated Python microservices execute computationally intensive signal processing and predictive modeling.

The developed pipeline features an automated, highly configurable preprocessing module that sequentially applies DC offset elimination, baseline wander correction, 50/60 Hz notch filtering, bandpass filtering, and advanced wavelet-based denoising. Following noise attenuation, the system extracts a high-dimensional feature matrix from 12-lead ECG signals, encompassing time-domain, frequency-domain, and clinically validated interval metrics.

For the predictive modeling phase, the system evaluates a heterogeneous suite of machine learning classifiers including Random Forest, XGBoost, LightGBM, and Multi-Layer Perceptron (MLP) architectures. To ensure clinical credibility and address the inherent opacity of these models, the framework integrates SHapley Additive Explanations (SHAP).

This inclusion provides granular, feature-level and lead-level interpretability, explicitly quantifying how individual physiological markers influence the model’s diagnostic output. Evaluated against the rigorously annotated PTB-XL dataset [18], the platform establishes a scalable, transparent, and highly interactive clinical decision support system.

1.1 Background and Motivation

The escalating global burden of cardiovascular pathology necessitates a paradigm shift from manual, labor-intensive diagnostic processes to automated, AI-augmented clinical workflows. While raw ECG signals embed profound physiological information, their extreme susceptibility to noise artifacts poses a persistent challenge. The primary motivation for this research stems from the critical need to democratize access to advanced cardiac diagnostics through scalable web technologies, while simultaneously satisfying the clinical imperative for model transparency and interpretability.

1.2 Problem Statement

While machine learning has accelerated the automated classification of cardiac arrhythmias, existing solutions exhibit profound systemic limitations. They predominantly operate as isolated, offline scripts rather than integrated clinical tools, suffer from extreme sensitivity to physiological noise, and operate as opaque "black boxes." A critical void exists for a unified, end-to-end framework capable of ingesting raw ECG signals, dynamically mitigating noise, extracting robust diagnostic features, generating highly accurate multi-class predictions, and providing mathematically sound, transparent explanations for its decisions in a real-time web environment.

1.3 Objectives

The primary objectives driving this research are to:

- Architect and deploy a robust, real-time ECG signal processing and analysis pipeline.
- Implement and optimize multi-stage digital filtering and wavelet-based denoising techniques.
- Extract a comprehensive, clinically relevant feature matrix from raw 12-lead ECG signals.
- Train, evaluate, and benchmark an ensemble of advanced machine learning classifiers.
- Integrate Explainable AI (SHAP) to decode model predictions into clinician-interpretable formats.
- Develop an intuitive, full-stack web application to facilitate seamless interaction, batch processing, and live data visualization.

1.4 Key Contributions

The core contributions of this thesis are summarized as follows:

- **End-to-End System Architecture:** Development of a full-stack, web-based platform bridging complex Python-based AI microservices with an interactive React frontend.
- **Advanced Preprocessing Pipeline:** Implementation of a dynamic, user-configurable signal denoising framework.

- **High-Fidelity Classification:** Empirical evaluation of boosting algorithms and neural networks, demonstrating superior predictive performance using XGBoost on complex ECG topologies.
- **Clinical Interpretability:** Deployment of SHAP to demystify algorithmic outputs, mapping predictive importance back to specific ECG leads and temporal features.
- **Empirical Validation:** Rigorous benchmarking utilizing the standardized PTB-XL clinical dataset [18].

1.5 Organization of the Thesis

To systematically present the research, methodology, and findings, the remainder of this thesis is structured as follows:

- **Chapter 2** presents a comprehensive literature review of existing ECG processing techniques and AI methodologies.
- **Chapter 3** details the proposed methodology, encompassing data acquisition, signal processing, and feature extraction.
- **Chapter 4** discusses the machine learning model development, evaluation metrics, and empirical results.
- **Chapter 5** explores the architecture and design of the web-based User Interface (UI).
- **Chapter 6** outlines the holistic system advantages and comparative performance benefits.
- **Chapter 7** elaborates on the system deployment strategy and infrastructure.
- **Chapter 8** summarizes the conclusions derived from this work.

- **Chapter 9** proposes avenues for future research, including real-time monitoring and mobile integration.

Chapter 2

Literature Review

The landscape of electrocardiogram (ECG) analysis has been profoundly transformed by the proliferation of large-scale, annotated datasets and concurrent breakthroughs in machine learning (ML) and deep learning (DL) paradigms. Extensive review literature underscores the paradigm shift towards artificial intelligence (AI) for automating diagnostic interpretations, particularly in enhancing the detection efficacy of myocardial infarction (MI) [1, 20]. The advent of comprehensive benchmark repositories, notably the PTB-XL dataset, has acted as a catalyst for this research trajectory, supplying high-fidelity, cardiologist-annotated 12-lead ECG signals essential for training robust and generalizable models [18].

Deep learning architectures have recently dominated the domain of automated ECG analysis. Frameworks leveraging Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and hybrid CNN-LSTM topologies have demonstrated exceptional diagnostic acuity in classifying complex cardiac anomalies [10, 17, 3]. Contemporary research is increasingly pivoting towards computational efficiency, advocating for lightweight, attention-driven models that facilitate real-time deployment in resource-constrained clinical environments without sacrificing predictive accuracy [9, 6].

Concurrently, traditional feature-centric machine learning methodologies maintain significant relevance, particularly when model interpretability is paramount. Standard algorithms, such as k-nearest neighbors and support vector machines, alongside ensemble classifiers, remain foundational to diagnostic workflows [huang2020adaboost, 2]. Within this spectrum, ensemble techniques specifically Random Forest, XGBoost, and other gradient boosting frameworks exhibit superior capability in navigating high-dimensional feature spaces and severe class imbalances, frequently yielding state-of-the-art predictive performance [16, 5, 21].

The efficacy of these predictive models is inextricably linked to rigorous signal preprocessing. Methodologies aimed at noise attenuation such as baseline wander correction, notch filtering for powerline interference, bandpass filtering, and sophisticated wavelet-based denoising are standard prerequisites to ensure signal fidelity prior to feature extraction [13]. Current literature strongly emphasizes the delicate balance required to suppress artifactual noise while strictly preserving clinically critical waveform morphologies.

Furthermore, the integration of Explainable Artificial Intelligence (XAI) has become an imperative in healthcare informatics, driven by the critical need for model transparency. Attribution frameworks, predominantly SHapley Additive Explanations (SHAP), dissect model predictions by quantifying the marginal contributions of distinct physiological features, thereby bridging the gap between algorithmic black boxes and clinical trust [lundberg2017shap].

Despite these substantive advancements, the prevailing literature frequently isolates individual pipeline components analyzing signal preprocessing or model architecture in silos. There remains a pronounced scarcity of holistic systems that synthesize real-time signal denoising, robust ensemble classification, and interactive visual analytics into a cohesive,

operational platform. This critical gap underpins the motivation for the proposed research, which architectures a unified, web-deployed framework seamlessly integrating advanced preprocessing, machine learning diagnostics, and XAI to deliver interpretable, real-time ECG analysis.

Chapter 3

Proposed Methodology

3.1 Overview

The methodology follows a structured, end to end pipeline for ECG based cardiac condition classification. The pipeline consists of six major stages: (1) signal acquisition and upload, (2) signal preprocessing and denoising, (3) feature extraction, (4) feature selection, (5) model training and classification, and (6) SHAP based explainability. Each stage is designed to be modular and configurable, allowing users to interact with and customize individual components through the web interface. A high level overview of the workflow is illustrated in Figure 3.1.

In addition to these six major stages, the web interface exposes a 7-stage signal processing viewer for transparency: the raw signal, intermediate denoising outputs, and a final raw vs denoised overlap check used to verify that denoising preserves clinically relevant morphology.

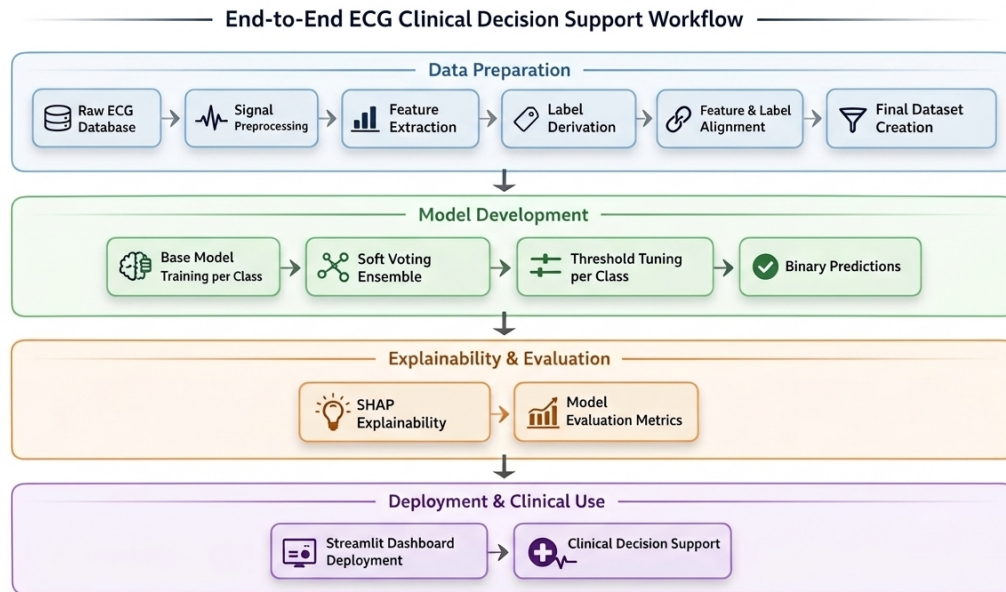


FIGURE 3.1: End to end workflow of the proposed ECG Analysis System.

3.2 Step 1: Data Acquisition and Signal Upload

The first step in the pipeline is the acquisition and ingestion of raw ECG signals into the system. The platform is designed to accept 12 lead ECG recordings in standard clinical formats, including WFDB (WaveForm DataBase) format as used in the PTB XL dataset [18]. Users upload ECG files through the React based web interface, which communicates with the Node.js backend via RESTful API calls.

Upon upload, the backend validates the file format and passes the signal to the Python processing engine. The system reads the signal data using the wfdb Python library, extracting the raw amplitude values along with metadata such as sampling frequency ($f_s = 500$ Hz for PTB-XL), number of leads, signal duration, and associated clinical labels.

The 12 lead ECG signals comprise the following leads: I, II, III, aVR, aVL, aVF (limb leads)

and V1V6 (precordial leads). Each lead captures electrical activity from a distinct anatomical perspective, providing complementary diagnostic information. The raw signals are stored as NumPy arrays of shape $(T \times 12)$, where T denotes the total number of time samples.

Dataset: The PTB-XL dataset contains 21,837 clinical 12 lead ECG records of 10 second duration from 18,885 patients, annotated with 71 ECG statements conforming to the SCP-ECG standard. For this study, the five superclass labels are used: Normal ECG (NORM), Myocardial Infarction (MI), ST/T-wave change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP).

3.3 Step 2: Signal Preprocessing and Denoising

Raw ECG signals are highly susceptible to various forms of noise and artifact contamination that arise during recording. To ensure reliable downstream analysis, a comprehensive multi stage preprocessing pipeline is applied to each lead independently before feature extraction. The pipeline consists of five sequential denoising operations, each targeting a specific type of noise or artifact. Users can toggle individual stages on or off through the web interface, enabling flexible experimentation.

For interpretability and quality assurance, the interface additionally renders a final overlap visualization (Stage 7 in the viewer) and computes feature-preservation metrics (correlation, energy retention, and peak retention) between the raw and denoised signals.

3.3.1 Stage 2.1: DC Offset Removal

DC offset (or DC bias) refers to a non zero mean value in the ECG signal, typically caused by electrode polarization or instrumentation issues. It manifests as a constant vertical

shift in the waveform that has no physiological significance but can interfere with subsequent filtering and feature extraction.

DC removal is performed by subtracting the signal mean from every sample:

$$x_{dc}(t) = x(t) - \frac{1}{T} \sum_{t=1}^T x(t) \quad (3.1)$$

where $x(t)$ is the raw signal at time sample t , and T is the total number of samples. This operation is computationally inexpensive and is always applied as the first stage, as it centers the signal around zero and improves the stability of subsequent filtering steps.

3.3.2 Stage 2.2: Baseline Wander Correction

Baseline wander (BW) is a low-frequency artifact (typically in the range of 0.05-0.5 Hz) caused by respiration, patient movement, and poor electrode contact. It appears as a slow undulation of the isoelectric baseline, making it difficult to accurately measure waveform amplitudes and clinical intervals such as the ST segment.

Baseline wander is removed using a high-pass Butterworth filter with a cutoff frequency of $f_c = 0.5$ Hz:

$$H_{HP}(s) = \frac{s^n}{s^n + \omega_c^n} \quad (3.2)$$

where n is the filter order (set to 4 in this work) and $\omega_c = 2\pi f_c$. To prevent phase distortion that would shift QRS complexes and alter interval measurements, zero-phase filtering is applied using `scipy.signal.filtfilt`, which processes the signal in both the forward and reverse directions:

$$x_{bw}(t) = \text{filtfilt}(b_{HP}, a_{HP}, x_{dc}(t)) \quad (3.3)$$

This effectively removes slow varying baseline drift while preserving the morphology of P waves, QRS complexes, and T waves.

3.3.3 Stage 2.3: Notch Filtering

Power line interference (PLI) is one of the most common sources of noise in ECG recordings. It appears as a sinusoidal artifact at the mains supply frequency 50 Hz in most countries (including India and Europe) or 60 Hz in North America along with its harmonics. PLI significantly contaminates the signal and must be removed before frequency-domain analysis.

A second order IIR notch filter (also called a band-stop filter) is designed using `scipy.signal.iirnotch` with quality factor $Q = 30$:

$$H_{notch}(z) = \frac{1 - 2 \cos(\omega_0)z^{-1} + z^{-2}}{1 - 2r \cos(\omega_0)z^{-1} + r^2z^{-2}} \quad (3.4)$$

where $\omega_0 = 2\pi f_0/f_s$ is the normalized notch frequency, $f_0 \in \{50, 60\}$ Hz, and $r = 1 - \pi f_0/(Q \cdot f_s)$ controls the sharpness of the notch. The filter attenuates the target frequency while leaving the surrounding spectrum relatively undisturbed. Zero phase filtering is again applied to preserve phase linearity.

3.3.4 Stage 2.4: Bandpass Filtering

Following removal of baseline wander and PLI, a bandpass filter is applied to retain only the frequency content clinically relevant to the ECG waveform. The diagnostically significant frequency range of the ECG is 0.540 Hz, which encompasses P waves, QRS complexes, and T-waves while excluding residual low-frequency drift and high frequency muscle noise (EMG artifacts).

A 4th-order Butterworth bandpass filter is constructed using `scipy.signal.butter`:

$$x_{bp}(t) = \text{filtfilt}(b_{BP}, a_{BP}, x_{bw}(t)) \quad (3.5)$$

where b_{BP} and a_{BP} are the numerator and denominator coefficients of the bandpass filter with lower cutoff $f_l = 0.5$ Hz and upper cutoff $f_h = 40$ Hz. The combination of high pass and bandpass filters in consecutive stages provides robust suppression of out of band noise components.

3.3.5 Stage 2.5: Wavelet Based Denoising

After conventional linear filtering, residual high frequency noise and transient artifacts may still remain in the signal. Wavelet based denoising is applied as an optional but highly effective final stage, particularly for suppressing Gaussian noise and short duration artifacts while preserving sharp features such as the QRS complex.

The Discrete Wavelet Transform (DWT) decomposes the signal into approximation and detail coefficients at multiple resolution levels:

$$x(t) = \sum_k c_{J,k} \phi_{J,k}(t) + \sum_{j=1}^J \sum_k d_{j,k} \psi_{j,k}(t) \quad (3.6)$$

where $c_{J,k}$ are the approximation coefficients at the coarsest level J , $d_{j,k}$ are the detail coefficients at level j , and ϕ, ψ are the scaling and wavelet mother functions, respectively. In this work, the db4 (Daubechies-4) wavelet is used with decomposition depth $J = 5$, as it closely resembles ECG morphology.

Soft thresholding is applied to the detail coefficients at each level to suppress noise while retaining signal structure:

$$\hat{d}_{j,k} = \text{sign}(d_{j,k}) \cdot \max(|d_{j,k}| - \lambda_j, 0) \quad (3.7)$$

The threshold λ_j at each level j is estimated using the universal threshold rule [13]:

$$\lambda_j = \sigma_j \sqrt{2 \ln N} \quad (3.8)$$

where N is the signal length and σ_j is the noise standard deviation estimated from the median absolute deviation of the detail coefficients at level j :

$$\sigma_j = \frac{\text{median}(|d_{j,k}|)}{0.6745} \quad (3.9)$$

The denoised signal is then reconstructed using the inverse DWT (IDWT) with the thresholded coefficients. This approach is particularly effective because ECG morphological features are concentrated in the lower frequency wavelet scales, while noise predominantly affects the higher frequency detail coefficients. The complete preprocessing pipeline is summarized in Algorithm 1.

Algorithm 1 ECG Signal Preprocessing Pipeline

Require: Raw ECG signal $x(t)$, sampling frequency f_s

Ensure: Denoised ECG signal $\hat{x}(t)$

- 1: $x \leftarrow x - \text{mean}(x)$ ▷ DC offset removal
 - 2: $x \leftarrow \text{filtfilt}(b_{HP}, a_{HP}, x)$ ▷ Baseline wander correction ($f_c = 0.5$ Hz)
 - 3: $x \leftarrow \text{filtfilt}(b_{notch}, a_{notch}, x)$ ▷ Notch filter at 50/60 Hz
 - 4: $x \leftarrow \text{filtfilt}(b_{BP}, a_{BP}, x)$ ▷ Bandpass filter (0.5–40 Hz)
 - 5: Compute DWT coefficients $\{c_J, d_1, \dots, d_J\}$ using db4 wavelet
 - 6: **for** $j = 1$ to J **do**
 - 7: $\sigma_j \leftarrow \text{median}(|d_j|) / 0.6745$
 - 8: $\lambda_j \leftarrow \sigma_j \sqrt{2 \ln N}$
 - 9: $\hat{d}_j \leftarrow \text{sign}(d_j) \cdot \max(|d_j| - \lambda_j, 0)$
 - 10: **end for**
 - 11: $\hat{x} \leftarrow \text{IDWT}(c_J, \hat{d}_1, \dots, \hat{d}_J)$ ▷ Signal reconstruction
 - 12: Compute preservation metrics between x_{raw} and $\hat{x}$ ▷ Correlation, energy retention,
peak retention
 - 13: **if** metrics below acceptance thresholds **then**
 - 14: Blend with raw: $\tilde{x}(t) \leftarrow (1 - \alpha) \hat{x}(t) + \alpha x_{raw}(t)$ ▷ Increase α until preserved
 - 15: $\hat{x} \leftarrow \tilde{x}$
 - 16: **end if**
 - 17: **return** $\hat{x}$
-

3.3.6 Stage 2.6: Feature Preservation Enforcement and Raw vs Denoised Overlay (Viewer Stage 7)

Denoising can unintentionally attenuate diagnostically relevant morphology (e.g., QRS sharpness or ST level). Therefore, after producing a denoised candidate signal, the system performs an automatic feature-preservation check against the original waveform and exposes the comparison to the user through an overlap plot.

Overlay (“imposing”) of raw and denoised signals

To visually validate preservation, the interface overlays the raw and denoised signals for a short fixed window ($N = 1000$ samples) on a shared vertical scale. Let $x_{raw}[n]$ and $x_{den}[n]$ denote the raw and denoised samples for the selected lead. The display window is:

$$\{x_{raw}[0], \dots, x_{raw}[N-1]\}, \quad \{x_{den}[0], \dots, x_{den}[N-1]\} \quad (3.10)$$

Both waveforms are scaled using the maximum absolute amplitude across the two windows to ensure a consistent overlay scale:

$$M = \max \left(\max_{0 \leq n < N} |x_{raw}[n]|, \max_{0 \leq n < N} |x_{den}[n]| \right), \quad s = \frac{1}{M} \quad (3.11)$$

The rendered overlay uses the same scale factor s for both signals, allowing direct visual comparison of morphology. In the viewer, raw is drawn in gray and denoised in purple.

Signal strength (amplitude range) stored for quick QA

Alongside the plot, a compact “signal strength” indicator is computed as the peak-to-peak amplitude range of the displayed denoised window:

$$A_{pp} = \max_{0 \leq n < N} x_{den}[n] - \min_{0 \leq n < N} x_{den}[n] \quad (\text{mV}) \quad (3.12)$$

This range value is stored and shown with the overlap visualization to provide a quick sanity check of amplitude scaling after preprocessing.

Feature-preservation metrics and enforcement

Beyond the visual overlay, three quantitative preservation metrics are computed over the common length L of the raw and denoised vectors:

- **Pearson correlation:** Measures morphological similarity.
- **Energy retention:** $ER(\%) = 100 \times \frac{\sum_{n=0}^{L-1} x_{den}[n]^2}{\sum_{n=0}^{L-1} x_{raw}[n]^2}$.
- **Peak retention:** Ratio of prominent peak counts (a lightweight proxy for QRS preservation).

If the denoised candidate does not meet minimum thresholds (correlation, energy retention, and peak retention), an enforcement step blends the denoised signal with the raw signal:

$$x_{out}[n] = (1 - \alpha) x_{den}[n] + \alpha x_{raw}[n], \quad \alpha \in [0, 1] \quad (3.13)$$

The smallest α that satisfies the acceptance criteria is selected. The final per-lead preservation metrics (including whether enforcement was applied and the selected blend factor α) are stored with the preprocessing output and displayed to the user.

3.4 Step 3: Feature Extraction

Following preprocessing, a rich set of features is extracted from each of the 12 ECG leads independently. The features are designed to capture a comprehensive description of cardiac activity across time domain, frequency domain, and clinically meaningful dimensions. In total, approximately 780 features are extracted across all 12 leads.

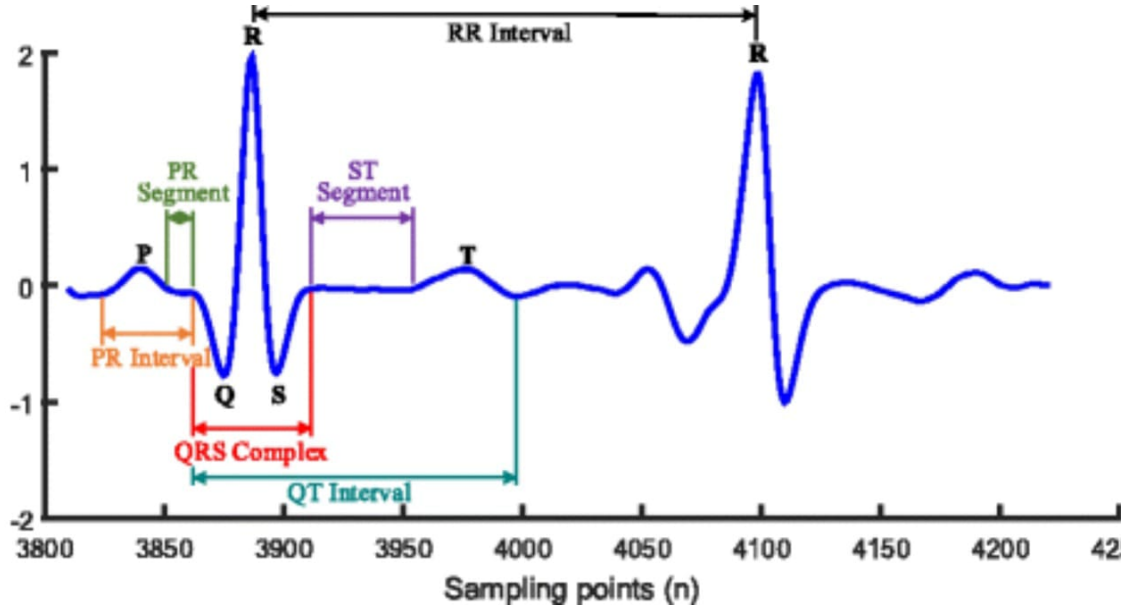


FIGURE 3.2: Annotated ECG waveform showing P-wave, QRS complex, T-wave, and clinical intervals (PR, QT, RR, ST).

3.4.1 QRS Detection and R-Peak Localization

Before computing beat level features, R-peaks (the dominant peaks of QRS complexes) must be accurately detected. The Pan-Tompkins algorithm [14] is employed for this purpose. The algorithm applies a sequence of signal processing operations differentiation, squaring, and moving window integration to produce a feature signal that accentuates QRS complexes, followed by adaptive thresholding to detect R-peak positions. Formally, let the filtered signal be $x_{bp}(t)$. The derivative is computed as:

$$y_1(t) = \frac{1}{8f_s} (-x_{bp}(t-2) - 2x_{bp}(t-1) + 2x_{bp}(t+1) + x_{bp}(t+2)) \quad (3.14)$$

The squared signal emphasizes large slopes:

$$y_2(t) = y_1(t)^2 \quad (3.15)$$

A moving window integral over a 150 ms window smooths the result to produce the final detection signal:

$$y_3(t) = \frac{1}{N_w} \sum_{k=t-N_w+1}^t y_2(k), \quad N_w = \lfloor 0.15f_s \rfloor \quad (3.16)$$

R-peaks are then identified at local maxima of $y_3(t)$ exceeding an adaptive threshold. The detected R peak indices $\{r_1, r_2, \dots, r_K\}$ enable beat level segmentation and the computation of inter beat intervals.

3.4.2 Time Domain Features

Time-domain features quantify the statistical and morphological characteristics of the raw ECG signal and its beat to beat variability:

- **Statistical moments:** Mean, variance, standard deviation, skewness, and kurtosis of the amplitude values for each lead.
- **RR intervals:** The time differences between consecutive R-peaks, $RR_k = r_{k+1} - r_k$, expressed in milliseconds.
- **Heart Rate (HR):** Estimated as $HR = 60/\overline{RR}$, where $\overline{RR}$ is the mean RR interval in seconds.
- **SDNN:** Standard deviation of all RR intervals, a global HRV measure: $SDNN = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (RR_k - \overline{RR})^2}$.

- **RMSSD:** Root mean square of successive differences, sensitive to short-term HRV:

$$\text{RMSSD} = \sqrt{\frac{1}{K-1} \sum_{k=1}^{K-1} (RR_{k+1} - RR_k)^2}.$$

- **pNN50:** Proportion of successive RR differences exceeding 50 ms.
- **Signal amplitude statistics:** Maximum, minimum, peak to peak amplitude, and zero crossing rate.

3.4.3 Frequency Domain Features

Frequency domain features are computed from the Power Spectral Density (PSD) of each lead, estimated using Welch's method [welch1967use]:

$$S_{xx}(f) = \frac{1}{KU} \left| \sum_{t=0}^{K-1} x(t) w(t) e^{-j2\pi ft/f_s} \right|^2 \quad (3.17)$$

where $w(t)$ is a Hann window, K is the segment length, and U is the window normalization factor. The following spectral features are extracted:

- **Band power:** Total power in the ULF (0–0.003 Hz), VLF (0.003–0.04 Hz), LF (0.04–0.15 Hz), and HF (0.15–0.4 Hz) bands.
- **LF/HF ratio:** Indicator of sympathovagal balance.
- **Spectral entropy:** Measures the flatness of the PSD, indicating signal regularity.
- **Dominant frequency:** The frequency at which the PSD is maximized.
- **Spectral centroid and bandwidth:** First and second moments of the power spectrum.

3.4.4 Clinical Interval Features

Clinically meaningful intervals are delineated using the detected R-peaks and morphological analysis of individual beats:

- **QRS duration:** Width of the QRS complex, measured as the time from Q-wave onset to S wave offset. Prolonged QRS (> 120 ms) may indicate bundle branch block.
- **PR interval:** Time from P-wave onset to QRS onset, reflecting atrioventricular (AV) conduction. Normal range: 120-200 ms.
- **QT interval:** Time from QRS onset to T wave offset, reflecting ventricular depolarization and repolarization. Corrected QT interval (QTc) is computed using Bazett's formula: $QTc = QT / \sqrt{RR}$.
- **ST segment level:** Amplitude of the signal 60–80 ms after the J-point (QRS offset), used to detect ST elevation or depression indicative of myocardial ischemia or infarction.
- **T-wave amplitude:** Peak amplitude of the T wave relative to the isoelectric baseline.
- **P wave features:** Duration and amplitude of the P-wave, important for detecting atrial abnormalities.

3.4.5 Heart Rate Variability (HRV) Features

HRV features quantify the variation in time between heartbeats and reflect the influence of the autonomic nervous system on cardiac function. Beyond the time domain HRV metrics listed above, the following nonlinear HRV features are also extracted:

- **Sample Entropy (SampEn):** Quantifies the regularity and unpredictability of the RR interval time series.

- **Approximate Entropy (ApEn):** A measure of irregularity; lower values indicate more regular patterns.
- **Poincaré plot descriptors (SD1, SD2):** Short and long term variability of RR intervals derived from the Poincaré scatter plot. $SD1 = RMSSD/\sqrt{2}$; $SD2 = \sqrt{2 SDNN^2 - SD1^2}$.

All features are extracted per lead and concatenated into a single feature vector $\mathbf{f} \in \mathbb{R}^{780}$ representing each ECG record.

3.5 Step 4: Feature Selection via Mutual Information

With approximately 780 features extracted from the 12 lead ECG, dimensionality reduction is essential to avoid overfitting, reduce computational cost, and improve model generalization. Feature selection is performed using the Mutual Information (MI) criterion, which measures the statistical dependence between each feature and the target labels in a non-parametric, model-agnostic manner.

For a multi label classification problem with C target classes, the average mutual information of feature X_j across all class labels $\{y_1, y_2, \dots, y_C\}$ is computed as:

$$\overline{MI}(X_j) = \frac{1}{C} \sum_{i=1}^C MI(X_j, y_i) \quad (3.18)$$

where the mutual information between feature X_j and binary label y_i is:

$$MI(X_j, y_i) = \sum_{x \in \mathcal{X}} \sum_{y \in \{0,1\}} p(x, y) \log \frac{p(x, y)}{p(x) p(y)} \quad (3.19)$$

Since the features are continuous, MI is estimated using the k -nearest neighbor estimator [ross2014mutual], implemented via `sklearn.feature_selection.mutual_info_classif`.

Features are ranked in descending order of $\overline{MI}(X_j)$, and the top $K = 40$ features are retained for model training. This aggressive reduction from ~ 780 to 40 features preserves the most discriminative information while substantially reducing the input dimensionality. The selected features tend to include QRS duration, ST-segment level, RR interval statistics, and lead specific spectral power, reflecting their high discriminative value for distinguishing cardiac conditions.

3.6 Step 5: Model Training and Classification

The ECG classification task is framed as a multi label binary classification problem, where each record may simultaneously belong to one or more cardiac condition classes. Four individual machine learning models are trained using the selected 40 dimensional feature vectors, and their performances are evaluated individually.

3.6.1 Data Splitting and Class Imbalance Handling

The PTB XL dataset is split into training (70%), validation (10%), and test (20%) sets using stratified random splitting to maintain class distribution across splits. Class imbalance particularly the over representation of the NORM class is addressed using the Synthetic Minority Over sampling Technique (SMOTE) [chawla2002smote], applied only to the training set to avoid data leakage.

3.6.2 Model 1: Random Forest (RF)

Random Forest is a bagging based ensemble of N_{trees} decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split. For a new input $\mathbf{f}$, the predicted class probability is the average of the probability estimates from all individual trees:

$$P_{RF}(y_i = 1 | \mathbf{f}) = \frac{1}{N_{trees}} \sum_{t=1}^{N_{trees}} p_t(y_i = 1 | \mathbf{f}) \quad (3.20)$$

Configuration: $N_{trees} = 500$, maximum depth = 20, minimum samples per leaf = 5. RF reduces variance through averaging and is robust to overfitting on high-dimensional data.

3.6.3 Model 2: XGBoost

XGBoost (Extreme Gradient Boosting) [5] is a gradient boosting framework that sequentially adds decision trees to minimize a regularized objective function:

$$\mathcal{L} = \sum_n l(y_n, \hat{y}_n) + \sum_t \Omega(f_t), \quad \Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|\mathbf{w}\|^2 \quad (3.21)$$

where l is the binary cross-entropy loss, T is the number of leaves, and $\mathbf{w}$ are the leaf weights. The regularization terms γ and λ prevent overfitting. New trees are fit to the negative gradient of the loss with respect to current predictions. Configuration: 300 estimators, learning rate $\eta = 0.05$, maximum depth = 6, subsample ratio = 0.8, column sample ratio = 0.8. XGBoost achieves the best individual performance among all models in this work.

3.6.4 Model 3: LightGBM

LightGBM [11] is a gradient boosting framework that uses histogram-based tree learning and leaf wise (best first) tree growth, enabling efficient training on large datasets. Unlike depth wise growth used in conventional GBDT, leaf wise growth selects the leaf with the greatest loss reduction at each step:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (3.22)$$

where G and H are the first and second-order gradient statistics. Configuration: 500 estimators, learning rate = 0.05, maximum leaves = 31, minimum child samples = 20.

3.6.5 Model 4: Multi-Layer Perceptron (MLP)

A fully connected feedforward neural network is trained as a complementary model to capture non-linear feature interactions that gradient boosting methods may miss.

The architecture consists of multiple hidden layers with ReLU activations and batch normalization. The transformation at each layer is defined as:

$$\mathbf{h}^{(l)} = \text{ReLU} \left(\text{BN} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right) \right) \quad (3.23)$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ represent the weights and biases of the l -th layer, $\text{BN}(\cdot)$ denotes batch normalization, and ReLU is the activation function.

Configuration: The network architecture follows layer sizes $[40 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow C]$, where C is the number of output classes. A dropout rate of 0.3 is applied to reduce overfitting. The model is trained using the Adam optimizer with a learning rate of 10^{-3}

and binary cross entropy loss. Training is performed for 100 epochs with early stopping based on validation loss.

3.6.6 Summary of Model Configurations

The model configurations are summarized in Table 3.1.

TABLE 3.1: Summary of Model Configurations

Model	Key Hyperparameters	Primary Advantage
Random Forest	350 trees, depth=20	Low variance, robust to noise
XGBoost	300 estimators, lr=0.05	Highest single model accuracy
LightGBM	400 estimators, lr=0.05	Fast training, efficient
MLP	3 layers [128,64,32]	Captures non linear patterns

3.7 Step 6: SHAP Based Explainability

To ensure transparency and clinical trustworthiness, the system integrates SHapley Additive exPlanations (SHAP) [lundberg2017shap] to provide post hoc, model agnostic explanations of individual predictions. SHAP is grounded in cooperative game theory and assigns each feature an importance value (Shapley value) that quantifies its marginal contribution to the model's output.

3.7.1 Theoretical Foundation

For a model f and an input $\mathbf{f} = (f_1, f_2, \dots, f_M)$, the SHAP explanation decomposes the prediction into an additive sum of feature contributions:

$$f(\mathbf{f}) = \phi_0 + \sum_{j=1}^M \phi_j \quad (3.24)$$

where $\phi_0 = \mathbb{E}[f(\mathbf{f})]$ is the base value (expected model output over the training set), and ϕ_j is the Shapley value for feature j , computed as:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|! (M - |S| - 1)!}{M!} [f(S \cup \{j\}) - f(S)] \quad (3.25)$$

Here, $\mathcal{F}$ is the set of all features, S is a coalition of features excluding j , and $f(S)$ denotes the model output with only the features in S observed (others set to their expected value). The Shapley value represents the average marginal contribution of feature j across all possible coalitions.

3.7.2 TreeSHAP for Gradient Boosting Models

For tree-based models (XGBoost, Random Forest, LightGBM), the TreeSHAP algorithm [lundberg2020local] is used, which computes exact Shapley values in polynomial time $\mathcal{O}(TLD^2)$, where T is the number of trees, L is the number of leaves, and D is the maximum depth. This makes SHAP computation computationally feasible for the full test set.

3.7.3 Explainability Outputs

The system generates the following SHAP based visualizations and outputs for each prediction:

- **Global feature importance:** A bar chart ranking features by mean absolute SHAP value $|\phi_j|$ across all test samples, revealing which features most consistently influence predictions.
- **SHAP summary (beeswarm) plot:** Displays the distribution of SHAP values for each feature, colored by feature value, illustrating both the direction and magnitude of feature effects.
- **Local explanation (force plot):** For a specific patient record, shows how each feature pushes the prediction above or below the base value ϕ_0 , enabling per patient interpretability.
- **Leadwise SHAP aggregation:** SHAP values for features belonging to the same lead are summed to produce a per lead importance score, revealing which ECG leads are most diagnostically informative for each condition. For example, leads V1-V4 are expected to show high importance for anterior MI detection.
- **SHAP interaction values:** Pairwise interaction effects between features are computed to identify synergistic feature combinations.

Chapter 4

Results

This chapter presents a systematic evaluation of the proposed ECG classification framework, benchmarking four prominent machine learning architectures: Random Forest (RF) [4], XGBoost (XGB) [5], LightGBM (LGBM) [11], and a Multi-Layer Perceptron (MLP) [8]. These models were selected for their proven efficacy in handling high-dimensional data and their capacity to capture the complex, non-linear dependencies inherent in digitized ECG signals.

To rigorously assess the impact of feature dimensionality on model generalization, experiments were conducted across two distinct feature subsets, comprising 40 and 70 selected features. Derived from comprehensive ECG signal processing pipelines, these subsets encapsulate critical morphological, temporal, and frequency-domain characteristics. This comparative methodology enables a granular assessment of how increasing feature richness influences both classification accuracy and system robustness.

The predictive performance was quantified using standard multi-label classification metrics [zhang2013review]. Specifically, the Micro F_1 -score measures the global prediction performance across all samples and labels, whereas the Macro F_1 -score evaluates performance uniformly across all classes, preventing majority-class dominance. Additionally, the Area Under the Receiver Operating Characteristic Curve (AUC) is computed in both

micro and macro aggregations to assess the models' discriminative capabilities. Finally, Hamming Loss is utilized to quantify the fraction of incorrectly predicted instance-label pairs, providing direct insight into label-wise prediction error. Together, these metrics ensure a balanced and exhaustive evaluation of the classification system.

4.1 Comparison Across Feature Sets

Figure 4.1 illustrates the comparative performance of the tree-based ensemble models (RF, XGB, and LGBM) utilizing the 40-feature and 70-feature subsets across the defined evaluation metrics.

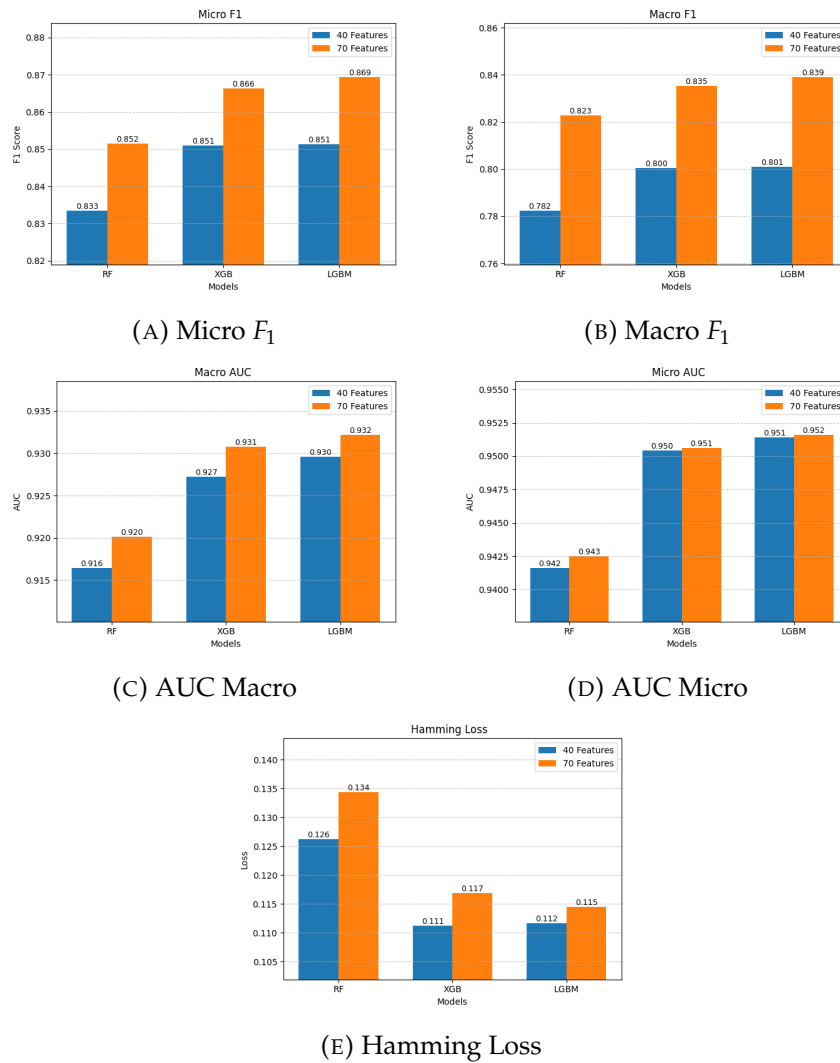


FIGURE 4.1: Comparative performance of RF, XGB, and LGBM architectures using 40 and 70 feature subsets.

The empirical results demonstrate that expanding the feature space from 40 to 70 yields consistent performance gains across all models and evaluation criteria.

Regarding the Micro F_1 -score, all architectures exhibit noticeable improvements with the richer feature representation. LightGBM achieves the highest overall performance (0.869), closely followed by XGBoost (0.866) and Random Forest (0.852). A parallel trend is observed in the Macro F_1 -score, where LightGBM again secures the leading performance (0.839), outperforming XGBoost (0.835) and Random Forest (0.823). This indicates that the gradient boosting frameworks are highly effective at maintaining classification balance across diverse and potentially imbalanced cardiac conditions.

In terms of class separability, both Micro and Macro AUC metrics improve with the 70-feature set. LightGBM yields peak AUC scores (Micro AUC ≈ 0.952 , Macro AUC ≈ 0.932), with XGBoost performing at a highly comparable level. This underscores the strong discriminative capability of gradient boosting algorithms in differentiating between complex multi-label diagnostic classes. Random Forest, while competitive, yields comparatively lower AUC values.

Furthermore, the Hamming Loss analysis corroborates these findings. Both XGBoost and LightGBM minimize this error metric more effectively than Random Forest, reflecting superior precision in their label-wise predictions.

4.2 MLP Model Performance

In addition to the ensemble models, a Multi-Layer Perceptron (MLP) was evaluated to benchmark the effectiveness of standard neural networks for this multi-label classification task.

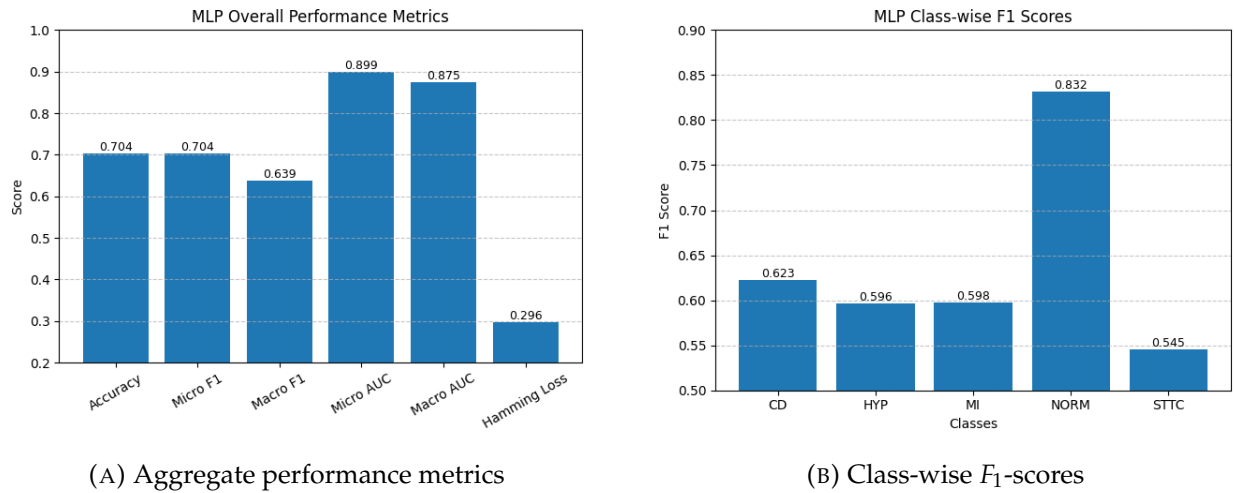


FIGURE 4.2: Performance evaluation of the Multi-Layer Perceptron (MLP) model.

The MLP model achieves a Micro F_1 -score of 0.7037 and a Macro F_1 -score of 0.6388, which are significantly lower than the benchmarks established by the tree-based models. While the Micro and Macro AUC values (0.8993 and 0.8746, respectively) suggest a moderate degree of discriminative capability, the elevated Hamming Loss (0.2963) reveals a pronounced vulnerability to label-wise prediction errors.

A granular, class-wise analysis (Figure 4.2b) indicates that the MLP performs optimally on the Normal (NORM) class ($F_1 \approx 0.83$). Conversely, significant performance degradation is observed in more complex diagnostic superclasses such as ST-T Change (STTC) and Conduction Disturbance (CD). This highlights the network's difficulty in independently abstracting the subtle morphological variations associated with these specific arrhythmias. Overall, while the MLP demonstrates foundational learning capabilities, it fundamentally underperforms the boosting frameworks within this tabular, feature-engineered paradigm.

4.3 Model Performance Analysis

Across the scope of this evaluation, gradient boosting methods (XGBoost and LightGBM) systematically outperform both Random Forest and the MLP network across all assessed metrics.

LightGBM demonstrates the most robust overall performance, securing the highest F_1 and AUC scores, establishing its superior capability in modeling complex, multi-label ECG distributions. XGBoost exhibits near-identical performance, confirming the general efficacy of gradient boosting for this application.

Conversely, Random Forest yields moderate results, while the MLP network exhibits distinct limitations. The neural network's lower F_1 -scores and higher Hamming Loss align with recent literature demonstrating that tree-based ensemble models consistently outperform unregularized deep neural networks on tabular datasets [grinsztajn2022tree]. The MLP's capacity to recognize broad patterns (evidenced by acceptable AUC scores) does not translate into the strict decision boundaries required for precise multi-label prediction.

4.4 Detailed Insights

The experimental results yield several critical insights regarding automated ECG analysis:

- **Impact of Feature Dimensionality:** Expanding the feature set significantly enhances model performance, confirming that rich morphological and spectral representations are vital for accurate cardiac condition classification.

- **Efficacy of Gradient Boosting:** XGBoost and LightGBM dominate the performance benchmarks due to their inherent structural advantages in modeling complex, non-linear feature interactions iteratively.
- **Deep Learning Limitations on Tabular Data:** The MLP's underperformance highlights the well-documented limitations of standard neural networks when applied to heterogeneous, tabular feature spaces without extensive regularization or massive scale [grinsztajn2022tree].
- **Discriminative Robustness:** The high AUC values achieved by the boosting models (> 0.93) denote excellent class separability, which is a fundamental requirement for reliable clinical diagnostic tools [18].
- **Precision vs. Recall Dynamics:** The observed variations in Hamming Loss emphasize the critical trade-off between sensitivity and precision, a persistent challenge in multi-label clinical datasets.

4.5 Conclusion

The experimental evaluation confirms that the proposed feature-based machine learning framework delivers robust and highly accurate multi-label ECG classification.

Among the investigated architectures, LightGBM and XGBoost emerge as the definitively superior models, maximizing predictive accuracy while minimizing label-wise error. While the MLP network provides baseline discriminative capabilities, it lacks the precision necessary for complex multi-label predictions within a structured feature setting.

Ultimately, these findings validate that gradient boosting techniques, when paired with comprehensive signal-processing feature extraction, offer a highly viable, computationally efficient solution for real-world ECG classification and automated clinical decision support.

Chapter 5

System Implementation and User Interface

The proposed methodology is instantiated as a comprehensive, web-based Clinical Decision Support System (CDSS) engineered for real-time electrocardiogram (ECG) analysis, interactive visualization, and the delivery of interpretable diagnostic insights. By seamlessly integrating robust data management, multi-stage signal processing, predictive model inference, and Explainable AI (XAI) modules into a unified interface, the platform ensures high clinical utility while retaining the granular flexibility required for advanced cardiological research [10].

5.1 Data Management and Visualization

The system architecture provides a highly interactive environment tailored for the efficient navigation and continuous monitoring of physiological records. A patient-centric data management module facilitates intuitive browsing and dynamic querying of records derived from the PTB-XL dataset [wagner2020ptb], concurrently maintaining a macroscopic overview of essential signal characteristics and metadata.



FIGURE 5.1: System snapshot displaying active signal parameters and dynamic configuration.

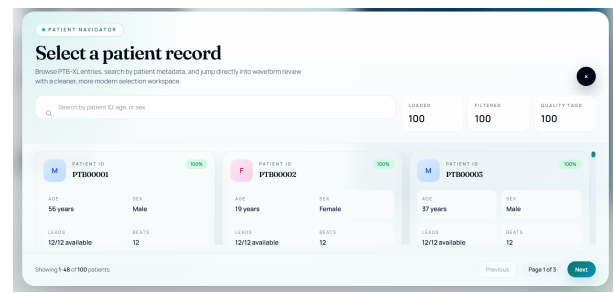


FIGURE 5.2: Patient navigation interface tailored for PTB-XL record selection and querying.

A dedicated multi-lead visualization workspace permits the simultaneous observation of the standard 12-lead ECG, preserving the critical spatial and temporal morphological relationships inherent in cardiac cycle interpretation. Furthermore, a highly interactive processing pipeline grants users unprecedented transparency into the sequential stages of signal preprocessing, filtering, and artifact reduction.

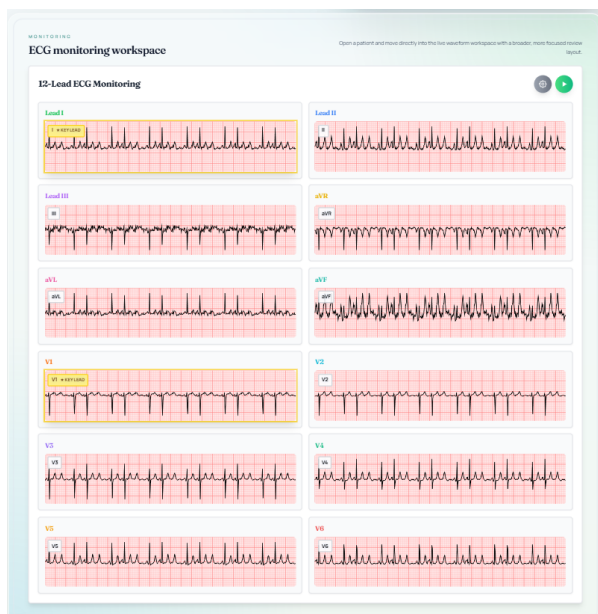


FIGURE 5.3: Comprehensive 12-lead ECG monitoring and visualization interface.

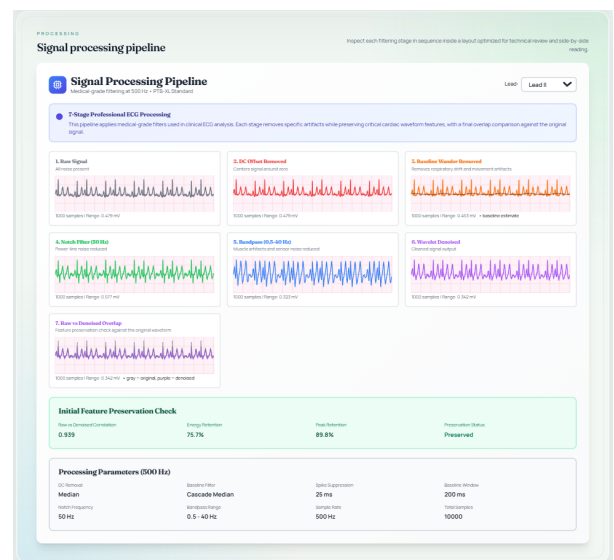


FIGURE 5.4: Multi-stage ECG signal processing and filtering pipeline.

5.2 Explainable Diagnostic Interface

The predictive outputs of the underlying deep learning models are surfaced via a specialized expert dashboard. This interface elegantly fuses multi-label classification probabilities with robust, interpretable evidence, empowering clinicians to transcend mere prediction viewing and actively scrutinize the model’s diagnostic rationale.

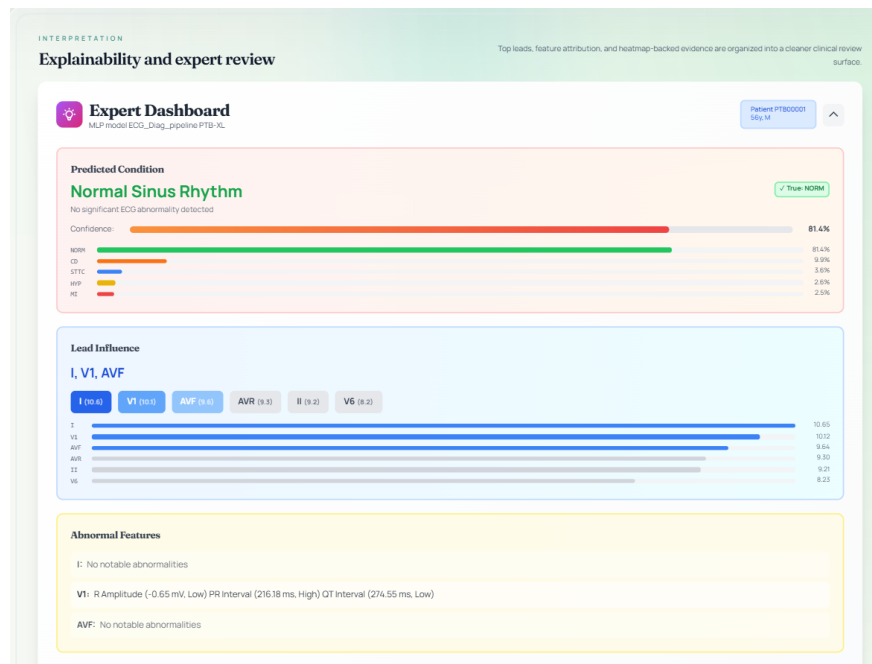


FIGURE 5.5: Expert dashboard integrating multi-label prediction outputs with lead-wise diagnostic importance.

To maximize diagnostic transparency, the framework supplies localized, feature-level explanations driven by XAI paradigms. Morphological highlighting isolates and emphasizes highly salient waveform segments (e.g., anomalous QRS complexes or ST-segment deviations), while high-resolution heatmaps project regions of dense model attention directly onto the time-series data.

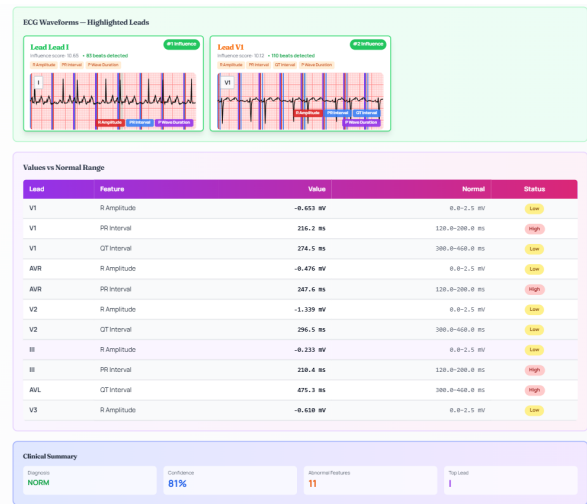


FIGURE 5.6: Morphological highlighting of clinically salient ECG segments.

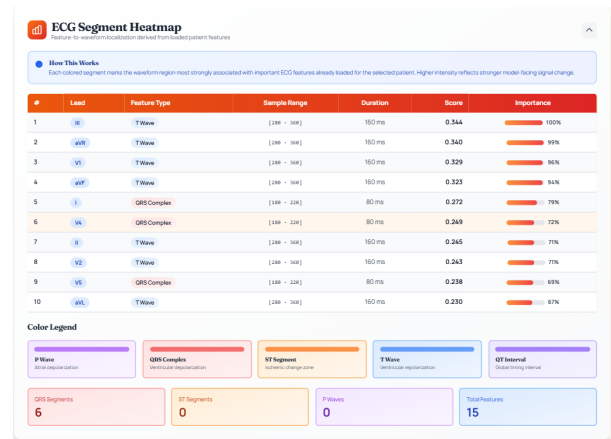


FIGURE 5.7: Segment-level importance heatmap illustrating model attention.

Granular feature interpretability is reinforced via structured clinical summaries and SHapley Additive exPlanations (SHAP)-based attribution rankings [12]. This allows for an exhaustive quantitative analysis of the specific morphological and demographic factors influencing the final predictive consensus.



FIGURE 5.8: Structured clinical feature summary table.

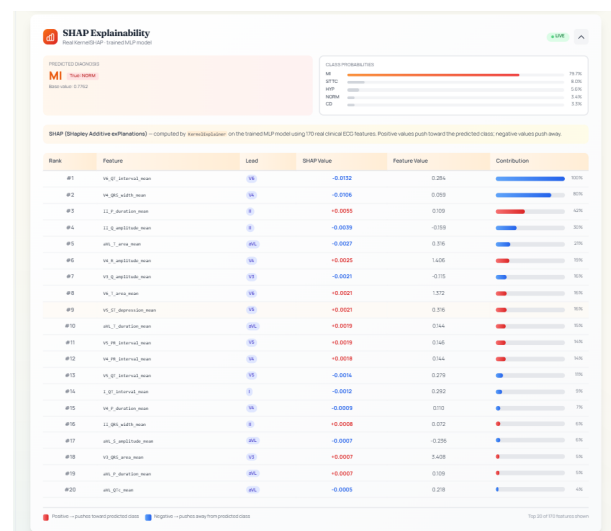


FIGURE 5.9: SHAP-based feature importance ranking for quantitative attribution.

5.3 Anatomical and Clinical Interpretation

Bridging the semantic gap between opaque algorithmic outputs and established clinical reasoning requires translating mathematical attributions into anatomical contexts. To this end, the system aggregates discrete feature contributions at the individual lead level. This novel aggregation allows cardiologists to contextualize model behavior with respect to the specific anatomical regions of the myocardium.

As illustrated in Figure 5.10, lead-wise SHAP aggregation dynamically delineates the relative predictive weighting of disparate ECG leads. This visualization is pivotal for clinical validation, allowing practitioners to verify whether the model correctly focuses on clinically corroborated regions such as prioritizing precordial leads for anterior septal abnormalities or inferior leads for ischemic events.

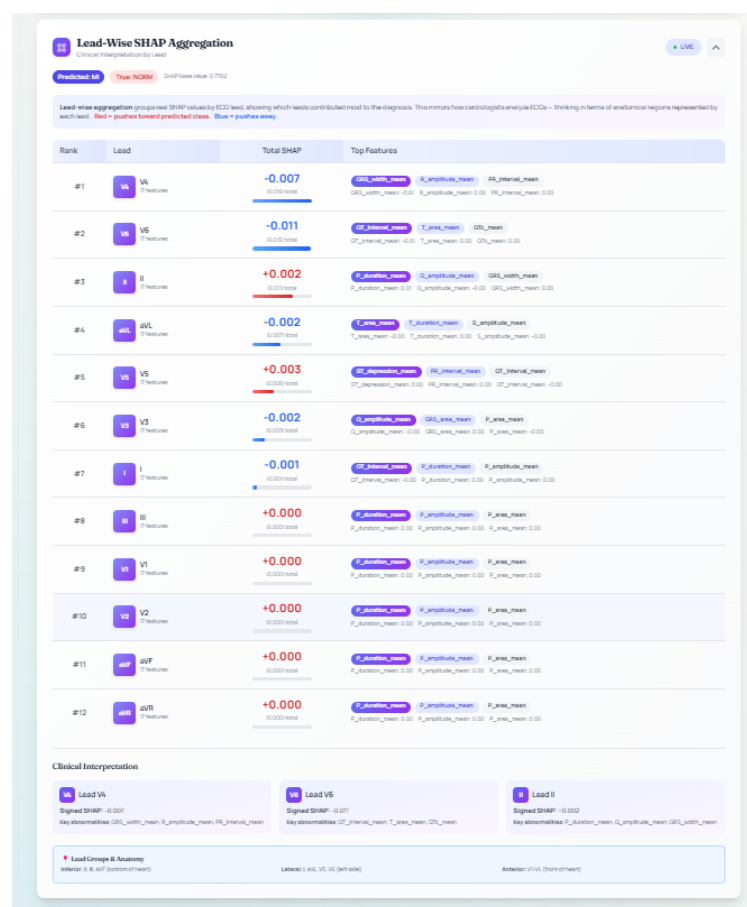


FIGURE 5.10: Lead-wise SHAP aggregation mapping algorithmic attention to anatomical regions.

Conclusively, the deployed system architectures establish a fully transparent, highly interpretable framework for automated ECG analysis. By synthesizing state-of-the-art signal processing, predictive machine learning, and clinically rigorous visual explanations within a cohesive user interface, the platform bridges the critical divide between artificial intelligence research and practical clinical implementation.

Chapter 6

System Advantages

The proposed ECG analysis platform offers several significant advantages over traditional systems by integrating advanced signal processing, machine learning, and modern web technologies into a unified framework. The key benefits of the system are outlined below:

- **End-to-End Automation:** The platform delivers a fully automated clinical decision support workflow, seamlessly bridging raw signal ingestion to diagnostic prediction and interpretation. By eliminating manual intervention across preprocessing, feature extraction, and classification, the system guarantees a reproducible and highly efficient diagnostic pipeline.
- **Real-Time Processing and Scalability:** Engineered for low-latency operation, the platform leverages a modern web-based architecture combining a React frontend, a Node.js backend, and optimized Python signal processing modules. This full-stack integration facilitates real-time data processing and live status updates, addressing the critical need for timely diagnostic feedback in fast-paced healthcare environments.
- **High Classification Accuracy:** Employing a robust ensemble of advanced machine learning models including Random Forest, XGBoost, LightGBM, and CatBoost the

system achieves exceptional predictive performance, highlighted by a micro F1-score of 0.85 across multiple diagnostic classes. Ensemble techniques, particularly XGBoost, excel in managing the high-dimensionality of 12-lead ECG features and effectively mitigate class imbalance [5, 16, 21].

- **Explainable Predictions:** To overcome the limitations of "black-box" artificial intelligence, the architecture natively integrates SHapley Additive Explanations (SHAP) [12]. By quantifying and visualizing the specific contributions of individual features and ECG leads, the system ensures clinical transparency, fostering clinician trust and actionable decision-making.
- **Flexible and Configurable Pipeline:** Designed with user agency in mind, the platform allows practitioners to dynamically control preprocessing parameters. Users can interactively toggle noise removal techniques such as baseline wander correction, notch filtering, and wavelet denoising [13] tailoring the analysis to specific clinical or research requirements.
- **Intelligent Feature Dimensionality Reduction:** Through mutual information-based feature selection, the system successfully condenses approximately 780 extracted features down to the 40 most diagnostically significant variables. This 95% reduction maintains high predictive power while ensuring per-lead physiological interpretability.
- **Comprehensive Visualization:** The intuitive interface provides immediate, clear visual feedback, juxtaposing raw and denoised ECG signals sourced from robust databases like PTB-XL [18]. Combined with visual SHAP summaries, these tools accelerate the clinician's ability to verify data quality and validate the AI's diagnostic reasoning.

Chapter 7

Clinical Deployment Workflow

The proposed system is engineered to integrate seamlessly into both clinical and research environments by providing a structured, user-friendly, and highly interactive analytical pipeline. The end-to-end operational workflow of the deployed ECG analysis platform is outlined below:

1. **Data Upload and Input Validation:** The workflow initiates when a clinician or technician uploads a 12-lead ECG recording via the React-based web interface. The system natively supports standard clinical data formats, such as the WFDB format [goldberger2000physiobank]. Upon submission, the backend executes strict validation protocols to confirm file integrity, compatible sampling frequencies (e.g., 500 Hz), and the presence of all 12 leads. Available metadata, such as signal duration and patient identifiers, are simultaneously parsed.
2. **Automated Signal Preprocessing:** Validated signals are subsequently routed through an automated, multi-stage preprocessing pipeline. This critical phase mitigates artifacts via a 0.05 Hz high-pass filter for baseline drift removal, a 100 Hz low-pass filter for high-frequency noise suppression, and notch filtering to eliminate 50/60 Hz powerline interference. Optional wavelet-based denoising is also available [13]. Real-time interactive visualizations juxtapose the raw and denoised

waveforms, empowering users to visually verify the effectiveness of the preprocessing stage.

3. **Feature Extraction and Dimensionality Reduction:** Following noise reduction, the system extracts approximately 780 comprehensive features across all 12 leads, capturing time-domain, spectral, and interval-based (QRS) characteristics. To optimize computational efficiency and mitigate the risk of overfitting, mutual information-based feature selection identifies the top 40 most diagnostically salient features. This rigorous dimensionality reduction preserves high discriminative power while ensuring lead-aware physiological relevance.
4. **Model Inference and Multi-Label Classification:** The refined 40-dimensional feature vector is processed by a highly tuned ensemble of machine learning models (incorporating Random Forest, XGBoost, LightGBM, and CatBoost) [16]. Utilizing soft voting and threshold tuning, the system performs multi-label classification to output precise probability scores across five primary diagnostic superclasses: Normal (NORM), Myocardial Infarction (MI), ST/T changes (STTC), Conduction Disorders (CD), and Hypertrophy (HYP).
5. **Results Visualization and Reporting:** The predicted classifications and their respective confidence probabilities are rendered on the frontend dashboard. Diagnostic results are structured via intuitive visual elements, such as probability bars and summary tables, facilitating rapid clinical interpretation. Furthermore, the architecture accommodates batch processing, enabling the simultaneous evaluation and comparison of multiple patient records in high-throughput settings.
6. **Explainability and Clinical Interpretation:** To establish algorithmic transparency, the platform integrates SHapley Additive Explanations (SHAP) [12]. The interface

delivers granular interpretability through three distinct lenses:

- **Global Feature Importance:** Highlights the most influential predictive variables across the entire training distribution.
- **Local Explanations:** Illustrates how specific features contribute to an individual patient's diagnostic prediction.
- **Lead-Wise Aggregation:** Traces algorithmic attention back to specific ECG leads, linking predictions to anatomical regions of the heart.

These explainability modules allow cardiologists to validate automated predictions against established clinical pathophysiology.

7. **User Interaction and Feedback Loop:** The application architecture affords users significant interactive control, such as dynamically toggling preprocessing filters or comparing individual base models. This operational flexibility serves both routine clinical screening and exploratory academic research. Future iterations of the platform aim to integrate clinician-in-the-loop feedback mechanisms to continuously refine model calibration and diagnostic accuracy.

Chapter 8

Conclusion

This research presents a comprehensive, web-based ECG analysis platform that seamlessly integrates advanced signal processing, machine learning, and explainable artificial intelligence (XAI) into a unified clinical decision support framework. The proposed architecture systematically addresses the primary bottlenecks in automated ECG diagnosis: noisy signal acquisition, high-dimensional data representation, and the critical need for algorithmic transparency.

A robust, multi-stage preprocessing pipeline was engineered to mitigate common artifacts such as baseline wander, powerline interference, and high-frequency noise while strictly preserving physiologically relevant waveform morphologies. Following signal refinement, a comprehensive set of approximately 780 features was extracted across all 12 leads, capturing time-domain, spectral, and interval-based characteristics. To optimize computational efficiency and model generalization, mutual information-based feature selection was deployed. This technique successfully condensed the feature space to the 40 most diagnostically salient variables (a 95% reduction) without compromising predictive power.

A rigorous comparative evaluation of multiple machine learning architectures including Random Forest, XGBoost, LightGBM, and a Multi-Layer Perceptron (MLP) was conducted

[16]. The optimized classification model, with XGBoost demonstrating superior predictive capability [5], exhibited exceptional performance on the benchmark PTB-XL dataset [18]. The system successfully categorized five distinct cardiac superclasses, achieving a micro F1-score of 0.85, a macro F1-score of 0.80, and AUC values exceeding 0.93. These metrics underscore the model's robustness and its ability to effectively handle the inherent class imbalances typical of clinical ECG data.

Beyond raw predictive performance, the integration of SHapley Additive Explanations (SHAP) bridges the critical gap between computational accuracy and clinical trust [12]. By providing transparent, feature-level and lead-wise insights into the model's predictive reasoning, the platform empowers cardiologists and clinicians to systematically verify automated diagnostic outputs against established physiological principles.

Furthermore, this study demonstrates the practical viability of deploying AI-driven diagnostics via a modern, full-stack web architecture. The synergy of a React-based interactive frontend, a scalable Node.js backend, and highly optimized Python analytical modules facilitates real-time processing and low-latency inference. This technical foundation ensures the system is highly accessible and scalable for both rigorous academic research and fast-paced clinical environments.

In conclusion, this work delivers an end-to-end paradigm for automated ECG analysis that harmonizes high classification performance with strict interpretability and usability. By translating advanced machine learning methodologies into a deployable, clinician-friendly tool, this system represents a substantive advancement toward reliable, transparent, and accessible AI solutions for cardiovascular disease diagnosis.

Chapter 9

Future Work

- **Real time ECG monitoring:** Integration with wearable ECG devices for continuous, streaming analysis.
- **Mobile application:** Development of a native iOS/Android app leveraging lightweight on device models.
- **Extended condition coverage:** Expanding the classification scope to rare arrhythmias and additional SCP-ECG labels beyond the five superclasses.
- **Federated learning:** Multi institutional model training while preserving patient data privacy.
- **Deep learning integration:** Investigating end to end CNN/Transformer models alongside feature based approaches for hybrid classification.

Appendix A

Additional Materials

This appendix provides supplementary materials for the PTB-XL ECG diagnosis project, including detailed algorithm pseudocode, implementation details, extra tables for reporting results, and reproducibility notes.

A.1 Repository Structure and Reproducibility

A.1.1 Top-level folders (implementation map)

The project is organized as follows:

- **scripts/**: data preparation, denoising, feature extraction, and model training.
- **server/**: backend services (denoising API and SHAP explainability API).
- **src/**: React web UI (visualization, interaction, and model outputs).
- **public/**: deployment-ready artifacts (processed JSON, trained model pickle, etc.).
- **archive/**: PTB-XL dataset directory (WFDB records + metadata CSV).
- **ECG_Diag_pipeline/artifacts/**: training outputs (metrics, checkpoints, derived artifacts).

A.1.2 Reproducibility commands

The following commands (as implemented in `package.json`) are the primary entry points:

- `npm run dev`: runs the web app + denoising server + SHAP server concurrently.
- `npm run denoise`: runs offline denoising (Python) to create `ptbxl_denoised/`.
- `npm run prepare-data`: prepares deployment artifacts under `public/`.

For training, the project includes Python scripts such as `scripts/train_resnet1d_v2.py` and `scripts/train_resnet1d_ptbxl.py`.

A.2 Denoising Pipeline (PTB-XL)

A.2.1 Five-stage denoising pipeline

Signals are denoised per lead using a multi-stage pipeline implemented in `scripts/denoise_ptbxl_data.py`. The default stages are:

1. DC offset removal (median or mean)
2. Baseline wander removal (cascade of median filters)
3. Notch filtering (powerline interference; typically 50 Hz)
4. Bandpass filtering (Butterworth; default 0.5–40 Hz)
5. Wavelet denoising (default: Daubechies db6, level 4, soft threshold)

A.2.2 Algorithm: ECG Signal Preprocessing Pipeline

Algorithm 2 presents the full ECG signal preprocessing pipeline as implemented in the denoising module.

Algorithm 2 ECG Signal Preprocessing Pipeline

Require: Raw ECG signal $x(t)$, sampling frequency f_s

Ensure: Denoised ECG signal $\hat{x}(t)$

- 1: $x \leftarrow x - \text{mean}(x)$ ▷ DC offset removal
 - 2: $x \leftarrow \text{filtfilt}(b_{HP}, a_{HP}, x)$ ▷ Baseline wander correction ($f_c = 0.5$ Hz)
 - 3: $x \leftarrow \text{filtfilt}(b_{notch}, a_{notch}, x)$ ▷ Notch filter at 50/60 Hz
 - 4: $x \leftarrow \text{filtfilt}(b_{BP}, a_{BP}, x)$ ▷ Bandpass filter (0.5–40 Hz)
 - 5: Compute DWT coefficients $\{c_J, d_1, \dots, d_J\}$ using db4 wavelet
 - 6: **for** $j = 1$ **to** J **do**
 - 7: $\sigma_j \leftarrow \text{median}(|d_j|) / 0.6745$
 - 8: $\lambda_j \leftarrow \sigma_j \sqrt{2 \ln N}$
 - 9: $\hat{d}_j \leftarrow \text{sign}(d_j) \cdot \max(|d_j| - \lambda_j, 0)$
 - 10: **end for**
 - 11: $\hat{x} \leftarrow \text{IDWT}(c_J, \hat{d}_1, \dots, \hat{d}_J)$ ▷ Signal reconstruction
 - 12: Compute preservation metrics between x_{raw} and $\hat{x}$ ▷ Correlation, energy retention, peak retention
 - 13: **if** metrics below acceptance thresholds **then**
 - 14: Blend with raw: $\tilde{x}(t) \leftarrow (1 - \alpha) \hat{x}(t) + \alpha x_{raw}(t)$ ▷ Increase α until preserved
 - 15: $\hat{x} \leftarrow \tilde{x}$
 - 16: **end if**
 - 17: **return** $\hat{x}$
-

A.2.3 Pseudocode: baseline wander removal (median cascade)

LISTING A.1: Baseline wander removal via median cascade.

```

1 # Input:  signal x[0..N-1], sampling rate fs
2 # Params: med_spike_ms, med_baseline_ms, smooth_baseline_ms
3
4 w_spike    = odd_int(med_spike_ms * fs / 1000)
5 w_baseline = odd_int(med_baseline_ms * fs / 1000)
6 w_smooth   = odd_int(smooth_baseline_ms * fs / 1000)
7
8 x_despike  = median_filter(x, kernel=w_spike)
9 baseline   = median_filter(x_despike, kernel=w_baseline)
10 baseline_s = median_filter(baseline, kernel=w_smooth)

```

```

11 y          = x - baseline_s
12 # Output: y

```

A.2.4 Feature-preservation enforcement (morphology guardrail)

To prevent over-smoothing and maintain ECG morphology, the denoised candidate signal is validated against the raw signal using three lightweight checks:

- Pearson correlation: $\rho(\mathbf{x}, \hat{\mathbf{x}}) \geq 0.85$
- Energy retention: $\frac{\|\hat{\mathbf{x}}\|_2^2}{\|\mathbf{x}\|_2^2} \times 100 \geq 75\%$
- Peak retention: $\frac{\text{\#peaks}(\hat{\mathbf{x}})}{\text{\#peaks}(\mathbf{x})} \times 100 \geq 70\%$

If constraints fail, the system blends the denoised candidate back toward the raw signal until constraints pass.

A.2.5 Pseudocode: enforce feature preservation by progressive blending

LISTING A.2: Progressive blending to enforce feature preservation.

```

1 # Input:  raw signal x, denoised candidate x_hat
2 # Params: blend_schedule = [0.15, 0.30, 0.45, 0.60, 0.75, 0.90,
3   1.00]
4
5 metrics = preservation_metrics(x, x_hat)
6 if metrics.pass:
7     return x_hat
8
9 for alpha in blend_schedule:
10     x_blend = (1 - alpha) * x_hat + alpha * x
11     metrics = preservation_metrics(x, x_blend)
12     if metrics.pass or alpha == 1.00:
13         return x_blend

```

A.2.6 Outputs produced by denoising

For each record and lead, the pipeline saves:

ptbxl_denoised/<record_path>/lead<k>_fs<fs>.npz

and aggregates run metadata in:

ptbxl_denoised/processed_metadata.csv
ptbxl_denoised/feature_preservation_summary.json

A.3 Feature Schema for Explainability (SHAP)

A.3.1 Clinical feature vector used for SHAP

The SHAP server (`server/shap_server.py`) constructs a clinical feature vector for each ECG record:

- 12 leads: {I, II, III, aVR, aVL, aVF, V1–V6}
- 17 clinical features per lead (e.g., QRS width, QT/QTc, ST elevation/depression)
- Total: $12 \times 17 = 204$ features

If the trained model expects a larger dimensionality (e.g., 240 features), the implementation pads the vector with zeros to match the expected input size.

A.3.2 Example feature naming convention

Features are flattened into names of the form `<Lead>_<FeatureKey>`:

LISTING A.3: Feature naming convention examples.

```
1 II_QRS_width_mean
2 V3_ST_elevation_mean
3 aVL_QTc_mean
```

A.3.3 SHAP configuration (implementation details)

The explainability service uses a kernel-based SHAP explainer:

- Background selection: random subset of patients and SHAP *k*-means summarization.
- Explainer: `shap.KernelExplainer(model.predict_proba, summary)`.
- Single-case computation: `nsamples=200, l1_reg="num_features(30)"`.
- Lead-level aggregation: sums absolute and signed SHAP values per lead.

A.4 Model Training Details (ResNet1D)

A.4.1 Label mapping: SCP codes to super-classes

For 5-class training, diagnostic SCP statements from PTB-XL are mapped to super-classes as shown in Table A.1.

TABLE A.1: SCP code to super-class mapping.

Superclass	Examples of mapped codes (not exhaustive)
NORM	NORM
MI	AMI, IMI, ILMI, LMI, PMI, ...
STTC	STD_, ISCA, ISCI, INVT, LNGQT, ...
CD	RBBB, LBBB, AVB, 1AVB, 2AVB, 3AVB, WPW, ...
HYP	LVH, RVH, LAO/LAE, RAO/RAE, ...

A.4.2 Input representation

Signals are loaded from PTB-XL WFDB records, arranged as a 12-lead matrix and resampled to 125 Hz to form a fixed-length input:

$$\mathbf{X} \in \mathbb{R}^{12 \times 1250}$$

corresponding to 10 seconds of ECG at 125 Hz.

A.4.3 Objective: multi-class focal loss with label smoothing

The model uses focal loss to reduce the impact of class imbalance:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{B} \sum_{i=1}^B (1 - p_{t,i})^{\gamma} \sum_{c=1}^C y_{i,c}^{(\text{smooth})} \log p_{i,c}$$

where $p_{t,i}$ is the probability assigned to the (smoothed) target, γ is the focusing parameter (e.g., 2.0–2.5), and $C = 5$.

A.4.4 Augmentation (signal-space)

The training scripts implement the following augmentation operations:

- Gaussian noise injection
- Amplitude scaling
- Baseline wander injection (low-frequency sinusoid)
- Random time shift
- Random lead dropout
- Segment masking (cutout)

A.4.5 Hybrid model: signals + extracted features

The hybrid trainer (`scripts/train_resnet1d_ptbxml.py`) concatenates a learned representation of the ECG signal with a selected subset of extracted tabular features. Feature selection uses mutual information to select $K = 300$ informative features. Missing values are handled using simple imputation and the features are scaled before model ingestion.

A.4.6 Training loop (high-level pseudocode)

LISTING A.4: High-level training loop.

```

1 # 1) Load PTB-XL metadata CSV
2 # 2) Map SCP codes -> 5-class label
3 # 3) For each ECG: load 12-lead waveform, resample to (12, 1250)
4 # 4) Load extracted feature table and align by record_id
5 # 5) Preprocess features: impute -> scale -> select K (mutual
   information)
6 # 6) Build dataset with augmentations + class-balanced sampling
7 # 7) Train with focal loss + label smoothing

```

```

8 # 8) Apply learning-rate scheduling (e.g., plateau or cosine
   restarts)
9 # 9) Optional: SWA accumulation near end of training
10 # 10) Evaluate with test-time augmentation (TTA) and report
    metrics

```

A.5 Hyperparameters (Reporting Tables)

A.5.1 ResNet1D v2 (signal-only) configuration

TABLE A.2: ResNet1D v2 signal-only hyperparameters.

Parameter	Value
Sampling rate (target)	125 Hz
Input length	1250 samples (10 s)
Leads	12
Classes	5 (CD, HYP, MI, NORM, STTC)
Batch size	64
Max epochs	150
Optimizer	Adam/AdamW (as configured in script)
Initial learning rate	5×10^{-4}
Min learning rate	5×10^{-6}
Weight decay	1×10^{-4}
Focal γ	2.0
Label smoothing	0.05

A.5.2 Hybrid ResNet1D-SE + selected features (key configuration)

TABLE A.3: Hybrid ResNet1D-SE hyperparameters.

Parameter	Value
Sampling rate (target)	125 Hz
Input length	1250 samples (10 s)
Selected tabular features	$K = 300$ (mutual information)
Batch size	64
Max epochs	200
Learning rate	3×10^{-4}
Weight decay	2×10^{-4}
Focal γ	2.5
Label smoothing	0.05
MixUp α	0.05
Early stopping monitor	macro-F1
SWA	enabled near final epochs (as configured)
TTA	enabled (multiple stochastic passes)

A.6 Backend APIs (for the Web App)

A.6.1 Denoising service (Node + Python)

The denoising API is implemented in `server/denoise-server.js`. Key routes:

- GET `/api/denoise/status`: returns completion state, processed record count, Python availability, and feature-preservation summary.
- POST `/api/denoise/start`: starts Python denoising for a given patient. Example JSON body: `{"patientId": "PTB00001", "targetFs": 500}`.
- GET `/api/denoise/output/<processId>`: returns stdout/stderr logs.
- POST `/api/denoise/stop/<processId>`: stops a running denoising process.

The Python denoising entry point accepts CLI arguments:

- `-patient-id` (optional; processes a single `ecg_id`)
- `-target-fs` (e.g., 500)
- `-out-dir` (default: `ptb1_denoised`)

A.6.2 SHAP explainability service (Flask)

The SHAP API is implemented in `server/shap_server.py`. Key routes:

- GET `/api/shap/status`: readiness, class list, cache size, etc.
- GET `/api/shap/<ecg_id>`: SHAP explanation for a single patient.
- GET `/api/shap/classes`: returns model class names.
- POST `/api/shap/batch`: batch explanations (list of ECG IDs).

A.7 Additional Experimental Results

A.7.1 Overall Performance Metrics

Based on the multi-label classification results, the ensemble model demonstrates robust generalization across the five diagnostic classes. Table A.4 summarizes the overall performance.

TABLE A.4: Overall performance metrics for 5-class PTB-XL classification.

Metric	Score / Value
Micro F1-score	0.85
Macro F1-score	0.80
NORM Class Recall	0.97

A.8 Short Code Excerpts

A.8.1 Denoising pipeline stages (excerpt)

LISTING A.5: Running the full denoising pipeline on a single lead.

```
1 results = run_pipeline(  
2     lead_signal,  
3     working_fs,  
4     dc_remove      = 'median',  
5     baseline_params = baseline_params,  
6     notch_params   = notch_params,  
7     bandpass_params = bandpass_params,  
8     wavelet_params  = wavelet_params,  
9 )  
10 cleaned_candidate = results['denoised']  
11 cleaned, metrics  = enforce_feature_preservation(lead_signal,  
    cleaned_candidate)
```

A.8.2 SHAP explanation call (excerpt)

LISTING A.6: Thread-safe SHAP value computation for a single patient.

```
1 with explainer_lock:  
2     shap_vals = explainer.shap_values(  
3         X_scaled,  
4         nsamples = 200,  
5         l1_reg    = "num_features(30)",  
6     )
```

End of Appendix A.

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